Framing the Strait of Hormuz Crisis on Social Media:
A Cross Platform Analysis of Reddit and YouTube

COSC2671 – Social Media and Network Analysis
Assignment 2 – Group 3 (PG)

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Abstract

This study analyses how Reddit and YouTube framed the 2026 Strait of Hormuz crisis. The corpus contains 43,298 processed English comments from 55 Reddit submissions, 291 YouTube videos, and daily FRED energy price context for 1 March–15 May 2026. We combine NLP with directed, weighted author reply networks: nodes are anonymised authors, and edges are replies from a user to the author being replied to. Both platforms were predominantly negative, but the cross platform sentiment gap is small. The dominant frames are geopolitical conflict, Hormuz/oil supply disruption, and market/presidential reaction. Reddit provides the stronger network, with 7,938 author nodes, 10,668 reply edges, 88.9% largest-component coverage, 28.2% reciprocity, and 55 cascades with median depth 9. YouTube adds scale, but its reply structure is shallow: only 12.36% of top-level comments receive any reply. Therefore, full networks are used for structure and diffusion analysis, while reduced communities are used only for interpretable sentiment and topic summaries.

1 Introduction

1.1 Problem formulation

The Strait of Hormuz is one of the world's most important petroleum chokepoints, so a credible military or shipping disruption can affect crude oil, refined products, aviation fuel expectations, and public discussion of market risk [1]. The 2026 crisis therefore provides a focused case for social media and network analysis: it is a geopolitical event with direct economic implications, but public interpretation is shaped through platform specific communities, reply structures, and discourse norms.

This project examines how that crisis was framed on Reddit and YouTube. The comparison is useful because the two platforms expose different forms of public discussion. Reddit provides deeper threaded interaction, making it stronger for reply network, influence, community, and cascade analysis. YouTube provides a larger broadcast oriented comment environment, where discussion is broader but reply structure is usually shallower. Studying both platforms allows to distinguish textual differences in sentiment and topics from structural differences in how interaction is organised.

1.2 Research question

> *How do Reddit and YouTube communities frame the Strait of Hormuz crisis, and how do sentiment, topics, influence, community structure, and diffusion differ across platforms?*

The analysis answers the research question using four main objectives:

1. describe where the Reddit, YouTube, and external price datasets come from, what

fields they contain, how they were preprocessed, and what their limitations are;

2. create directed and weighted author reply networks to study interaction structure, influence, communities, and robustness;

3. identify the main sentiment patterns and topic frames;

4. bring together the NLP and network results to compare Reddit and YouTube at both platform and community levels, noting cases where the data or platform structure limits interpretation.

2 Data collection and preprocessing

2.1 Collection design and provenance

The dataset combines Reddit threads, YouTube comments, and external energy price context for 1 March–15 May 2026. The collection was designed as a topic focused sample of discussion about the Strait of Hormuz crisis as a geopolitical and jet-fuel market risk.

Table 1 summarises the collection design and final processed corpus. Reddit contributes a smaller but more structured discussion sample: 55 submissions and 17,162 final comments from eight subreddits. YouTube contributes greater scale, with 291 videos and 26,136 final comments, but its collection is organised around video search results. FRED contributes only external price context.

Reddit was collected from public thread JSON endpoints on `www.reddit.com`. The sampling frame used eight subreddits: `worldnews`, `geopolitics`, `oil`, `energy`, `wallstreetbets`, `stocks`, `investing`, and `Economics`. Within each subreddit, four searches were unioned: `hormuz`, `iranoil`, `jetfuel`, and `persiangulf`. Candidate threads were filtered to the project window, ranked by engagement, and hydrated using Reddit's threaded JSON structure with `old` comment order.

YouTube was collected with the YouTube Data API v3 using six topical queries: Strait of Hormuz oil crisis; Hormuz Iran tanker; jet fuel price spike 2026; Brent crude Hormuz; Persian Gulf oil shipping; and OPEC Hormuz response. The collector retrieved up to 50 videos per query and up to 500 available comments or replies per video. External context was downloaded from FRED for daily U.S. Gulf Coast jet fuel, Brent crude, and WTI crude series. These price series are used only for temporal context, not as causal evidence.

Table 1: Collection provenance and final processed corpus.

Source	Collection frame	Raw collection	Comments	Authors
Reddit	8 subreddits; 4 keyword searches	55 submissions; 18,294 comments	17,162	10,795
YouTube	6 API search queries; up to 50 videos/query; up to 500 comments/video	291 videos; 30,878 API comments	26,136	19,676
FRED	Jet fuel, Brent, and WTI	3 daily price series	–	–

2.2 Preprocessing

The raw Reddit and YouTube data were first transformed into comment level tables. YouTube videos, comments, and replies were extracted from the API JSON, while Reddit threads were unpacked from their nested structure. This ensured that each comment retained the relevant metadata, such as video or submission information, parent comment details, reply depth, score, and author information. The same preprocessing pipeline was then applied to both platforms, except for platform specific metadata fields, domain stopwords, and a small Reddit filter for moderation.

Initial exploration before cleaning: Before applying any cleaning or preprocessing steps, the raw datasets were first explored to understand their overall size, structure, and potential sources of noise. This included checking date coverage, missing or empty comments, comment lengths, reply structures, and activity distribution across subreddits, videos, and channels. The purpose of this step was to identify data quality issues before they were removed or transformed, making the later preprocessing decisions more transparent and auditable. For Reddit, the raw inspection confirmed a nested discussion structure with 55 submissions, maximum observed reply depth of 9 after preprocessing, and substantial variation by subreddit. For YouTube, it confirmed a larger but shallower comment environment: the final cleaned table contains 18,012 top-level comments and 8,124 replies, meaning replies make up 31.1% of processed YouTube comments. The initial inspection also motivated the attrition reporting below, so that removed language, duplicate, empty, and bot template comments are visible.

Author anonymisation: Raw public author names were not used in the processed analysis tables. Instead, each author was converted to a stable 16-character HMAC-SHA256 hash [2]. This preserves the ability to count unique authors, construct reply networks, detect repeated participation, and join comments to communities, while avoiding raw usernames. Reddit also stores `parent_author_hash`, because the Reddit reply tree is later converted into directed author to author edges. The final processed corpus contains 10,795 Reddit author hashes and 19,676 YouTube author hashes.

Language detection and filtering: Language filtering was applied after flattening but before text modelling. The notebooks use a deterministic `langdetect` setup with a short text guard: very short Latin script comments with less than 50 characters are treated conservatively as English, while longer non-Latin comments are passed through language detection `langdetect`. This adjustment was required because language detection methods tend to be unreliable for short social media texts, where limited linguistic context and informal phrasing can produce inconsistent classifications [3].

Comments classified as non English were removed to keep tokenisation, lemmatisation, sentiment models, and topic models in a consistent language space. This removed 183 Reddit comments, leaving 18,111 English comments, and 1,904 YouTube comments, leaving 27,538 English ones. The larger YouTube loss is consistent with YouTube's broader international audience.

Platform specific noise filtering: Before general cleaning, Reddit comments were passed through an additional filter to remove obvious automated moderation messages. These included automatic removal notices, rule reminders, report summaries, and repeated subreddit specific template comments. The filter used known moderation account names and recurring template phrases, removing 61 Reddit comments. No equivalent filter was applied to YouTube because it did not contain the same structure.

Text cleaning: Two different cleaning techniques were used for different purposes. Firstly, `raw_text` preserves the original public comment for audit. Next, `clean_text` is the main modelling text: it normalises dotted acronyms such as `U.S.` and `U.K.` to `usa` and `uk` for easier tokenization, strips URLs, HTML entities, HTML tags, mentions, timestamps, zero-width characters, and emojis, normalises smart quotes and dashes, collapses repeated punctuation, lowercases text, and normalises whitespace. Nevertheless, `vader_text` is cleaned more lightly so that capitalisation and punctuation remain available as sentiment-intensity cues for VADER. After aggressive cleaning, 36 Reddit comments and 396 YouTube comments became empty and were removed.

Duplicate and near-duplicate filtering: Duplicate filtering used two passes. The first removed exact duplicate `clean_text` values, keeping the first occurrence. This removed 671 Reddit comments and 646 YouTube comments. The second used a near-duplicate key that lowercased text, removed digits and punctuation, and collapsed whitespace. This removed a further 96 Reddit comments and 215 YouTube comments. This decision reduces spam, boilerplate, and copied comments before frequency, sentiment, and topic analysis, but it is also a limitation because repeated slogans can be socially meaningful.

Tokenisation, stopwords, and lemmatisation: The notebooks first inspected token frequencies with only base stopwords removed, then added data driven domain stopwords. We manually inspected the 30 most frequent words and took the ones that appear often and are not relevant for the analysis. Substantive project terms such as `iran`, `oil`, `hormuz`, `jet`, `fuel`, and `war` were retained.

The Reddit list contains 50 terms, including generic fillers and Reddit interface words such as `reddit`, `subreddit`, `thread`, `post`, `op`, `edit`, `deleted`, and `removed`. The YouTube list contains 49 terms, including interface and engagement words such as `video`, `videos`, `channel`, `subscribe`, `watch`, and `watching`. Comments with zero useful tokens after this final pass were removed: 85 from Reddit and 145 from YouTube.

Table 2: Comment attrition during preprocessing.

Stage	Reddit kept	Reddit removed	Youtube kept	Youtube removed
Flattened/windowed raw data	18,294	20	29,442	1,436
English-language only	18,111	183	27,538	1,904
Moderation/bot filter	18,050	61	–	–
Non-empty cleaned text	18,014	36	27,142	396
Exact duplicates removed	17,343	671	26,496	646
Near-duplicates removed	17,247	96	26,281	215
Final non-empty tokens	17,162	85	26,136	145

Table 2 summarises how many comments remained or were removed after each stage. It shows that both datasets were reduced gradually. For Reddit, the sample decreased from 18,294 comments after flattening and window filtering to 17,162 final tokenised comments. For YouTube, the sample decreased from 29,442 comments to 26,136 final tokenised comments.

The largest reductions came from language filtering and duplicate removal, especially for YouTube, where 1,904 non-English comments and 646 exact duplicates were removed. Reddit also had an additional moderation filter, which removed 61 comments that were unlikely to represent ordinary user discussion.

Table 3: Final processed corpus used for downstream analysis.

Platform	Comments	Authors	Scopes	Active dates	Median tokens
Reddit	17,162	10,795	8 subreddits	75	8
YouTube	26,136	19,676	114 channels	76	7
Total	**43,298**	**30,471**	–	–	–

Table 3 summarises the final processed corpus used for downstream analysis. YouTube contributes more comments and authors, while Reddit contributes a smaller but deeper threaded corpus. The platforms have similar date coverage and median comment length, which supports comparison, but their interaction structures remain different: Reddit activity is grouped by subreddit and thread, while YouTube activity is spread across channels, videos, top-level comments, and shallow replies.

3 Methods

3.1 Analytical workflow

The analysis starts from the processed Reddit and YouTube comment tables described in Section 2. The methods combine NLP and network analysis: sentiment analysis measures the tone of discussion, topic modelling identifies the main discourse frames, reply networks describe interaction structure, community analysis links network groups back to sentiment and topics, and diffusion analysis examines how discussion develops through reply trees and engagement patterns.

3.2 Sentiment analysis

Sentiment was measured with three methods. First, an opinion-word baseline counts positive and negative words using the Bing Liu lexicons. The score is the number of positive tokens minus negative tokens, with positive, negative, and zero scores mapped to positive, negative, and neutral labels. This method is transparent but limited because it cannot handle context, negation, sarcasm, or political language.

Second, VADER [4] was applied to the lightly cleaned `vader_text` field. VADER is useful for social-media text because it uses punctuation, capitalisation, and intensifiers when computing its compound score. Comments were labelled positive when the compound score was at least 0.05, negative when it was at most -0.05, and neutral otherwise.

The main method is a transformer sentiment classifier, `cardiffnlp/twitter-roberta-base-sentiment-latest`, based on TimeLMs/twitter-RoBERTa [5]. For each comment, the model returns negative, neutral, and positive probabilities. The final label is the highest probability class, and the signed sentiment score is:

$$\text{sentiment score} = P(\text{positive}) - P(\text{negative}).$$

This score is used for platform level sentiment, topic level sentiment, community sentiment, and daily sentiment trends.

The transformer is used for headline results because it gives full corpus coverage and uses more context than lexicon methods. VADER and opinion-word scores are retained as baselines, and the methods are compared using label distributions, agreement rates, confusion matrices, and Cohen's kappa. Platform differences are interpreted with effect sizes and confidence intervals, not p-values alone.

3.3 Topic modelling

Topic modelling was used to identify broad discourse frames. The main method is Latent Dirichlet Allocation (LDA) [6], fitted separately for Reddit and YouTube. Separate models were used because the two platforms have different vocabularies, audiences, and interaction structures; forcing both into one topic model would hide platform specific framing.

The topic corpus uses non-empty `lemma_text`. Text was count-vectorised with unigrams and bigrams, a maximum vocabulary of 5,000 features, `min_df=5`, and `max_df=0.95`. LDA was fitted on count vectors because it models word counts rather than TF–IDF weights. Candidate topic counts were checked, and three topics per platform were selected because this gave interpretable, reportable frames without producing fragile micro-topics.

Each comment is assigned a dominant topic and topic probability. Topics are labelled manually using top terms, representative comments, platform scope, and mean sentiment. The Reddit topics capture market/presidential reaction, Iran–U.S. geopolitical conflict, and Hormuz/oil-market disruption. The YouTube topics capture military escalation, global power politics, and oil/gas supply. These labels are analyst interpretations of statistical topics, not discovered ground truth.

Topic quality was checked using UMass coherence [7], dominant-topic probabilities, and seed-stability tests based on top-term Jaccard overlap. A combined topic model is used only for cross platform association tests; interpretation relies on the platform specific LDA models.

3.4 Reply networks, influence, and communities

For each platform, an author reply network was constructed. Nodes are anonymised authors. Directed edges run from the replying author to the author being replied to. The graph is directed because reply direction identifies interaction roles, and weighted because the same pair of authors can interact multiple times. Repeated replies between the same source and target are collapsed into a single weighted edge. Self-replies, missing authors, and anonymous endpoints are removed.

Reddit edges are built from comment parent links and `parent_author_hash`, giving a deeper author to author reply graph. YouTube edges are built from replies to top level comments, so the YouTube graph is interpreted as a shallow reply participant network.

Network analysis was implemented with NetworkX [8]. The structural measures include nodes, weighted edges, density, largest weakly connected component, reciprocity, clustering, transitivity, and component structure.

Influence was measured using weighted PageRank, in-degree, out-degree, betweenness, eigenvector centrality, and Katz centrality. PageRank is the headline influence measure because it gives more weight to authors who receive replies from other important or active authors. Influence is analysed in terms of concentration, measured through PageRank Gini and the PageRank share held by the top 1% of users, rather than through definitive claims about the "most influential" individuals.

Communities were detected with Louvain on the undirected weighted projection of each reply network [9]. The undirected projection is used because the aim is to find dense interaction groups, while edge weights preserve repeated reply strength. Community results are interpreted only after quality checks, including community size, comment coverage, author coverage, scope purity, and robustness.

To connect network structure back to text, author level Louvain communities are joined to comment level sentiment and topic outputs. For each reportable community, the analysis summarises comment count, author count, dominant subreddit or channel, mean sentiment, sentiment-label shares, dominant topic, topic mixture, topic entropy, and representative comments. Author normalised sentiment and bootstrap intervals are used to reduce the effect of highly prolific users.

Robustness is assessed in two ways. PageRank stability is tested by bootstrap edge resampling and Kendall's tau for top ranked authors. Louvain stability is tested across random seeds using Adjusted Rand Index [10]. These checks prevent the report from overclaiming exact influence rankings or exact community boundaries.

3.5 Diffusion and engagement

Diffusion is analysed differently by platform because Reddit and YouTube expose different reply structures. Reddit supports nested threads, so each Reddit submission is treated as a cascade. For each cascade, the analysis measures size, maximum depth, mean depth, branching, leaf share, bushiness, unique users, concentration in the most active user, time to first reply, time to peak activity, and time to 50% of comments.

The Reddit cascade table is joined with sentiment, topic, and network outputs. This allows cascade behaviour to be compared by sentiment, dominant topic, and early participation by high-PageRank authors. Kruskal–Wallis tests are used for subreddit-level differences in cascade size and depth, while Spearman correlations examine associations between cascade size, depth, timing, sentiment, topic entropy, and early influential participation.

YouTube is not treated as an equivalent cascade network because replies mostly attach to top-level comments. Instead, YouTube diffusion is reported as an engagement analysis. For each top-level comment, the analysis measures whether it received replies, how many replies it received, and how quickly the first reply arrived. This allows YouTube to be compared as a shallow engagement environment without forcing it into Reddit's cascade structure.

As a supplementary network-diffusion check, the Reddit reply graph is also tested with an Independent Cascade simulation. Seed sets based on PageRank, betweenness, and random selection are compared across different seed budgets.

3.6 Platform comparison and external context

The final comparison stage combines sentiment, topic, network, community, diffusion, and external price outputs into a platform-level synthesis. Sentiment differences are evaluated with mean differences, Welch tests, Mann–Whitney rank-biserial effect sizes, bootstrap confidence intervals, and chi-square/Cramer's V tests on sentiment labels. Topic platform association is tested in the combined topic space using chi-square and Cramer's V.

External context is added by aligning daily discussion volume and daily negative sentiment share with FRED jet fuel, Brent, and WTI price series. Correlations are computed on raw and first-differenced series across short lags. These results are treated as exploratory temporal context only: the window is short, multiple lags are tested, and correlation cannot show whether market prices caused discussion or discussion anticipated price changes.

4 NLP Results: Sentiment and Topics

4.1 Sentiment and crisis tone

The sentiment results show that both Reddit and YouTube discussed the Strait of Hormuz crisis in a predominantly negative tone. As Figure 1 shows, the two platform distributions are similar: Reddit is slightly more negative on average, but the difference is small relative to the shared negative crisis tone.

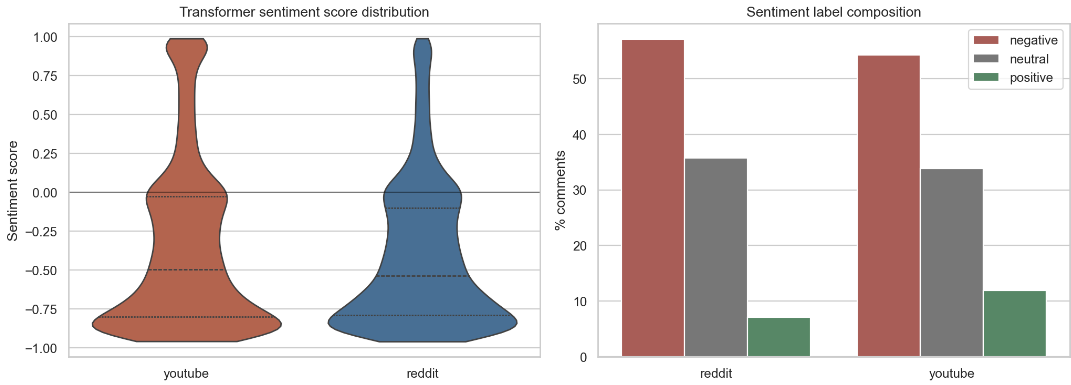

Figure 1: Transformer sentiment by platform. Both platforms are negative; Reddit is slightly more negative on average, but the distributional effect is small.

Table 4 summarises the main sentiment and lexical frame results. Reddit has a higher negative share and a more negative mean score, but the effect size is small: the Reddit–YouTube mean difference is -0.061 and the rank-biserial effect size is only -0.035. This means the platform difference is statistically detectable, but it should not be interpreted as a large emotional separation between audiences.

Table 4: Sentiment and crisis-frame indicators by platform.

Platform	Neg. %	Mean	Uncert./fear	Market spec.	Oil supply	Shipping
Reddit	57.1%	-0.406	21.45%	11.43%	9.20%	6.47%
YouTube	54.3%	-0.344	13.32%	7.58%	11.98%	8.82%

The frame indicators help explain the platform difference. Reddit's negativity is more strongly connected to uncertainty and market speculation, which fits the presence of finance and investing subreddits in the sample. YouTube has relatively more oil-supply and shipping/tanker language, suggesting a stronger connection to physical disruption and energy-supply narratives. This supports a more nuanced interpretation than simply saying one platform is more negative. Both are negative, but the tone is attached to slightly different crisis frames.

The comparison between sentiment methods also supports caution. VADER and the transformer agree on only 56.16% of comments, with Cohen's $\kappa = 0.327$. Opinion-word counting and VADER agree on 60.04% of comments, while opinion-word counting and the transformer agree on 54.84%. These moderate agreement levels mean that exact sentiment percentages should not be treated as ground truth. The robust conclusion is broader and the collected discourse is crisis-oriented and predominantly negative across methods.

4.2 Topic frames

The platform-specific LDA models identify three broad discourse frames per platform. Table 5 presents the topic prevalence and mean sentiment.

Table 5: Platform-specific LDA topic prevalence and sentiment.

Platform	Topic frame	Comments	Share	Mean sentiment
Reddit	Trump, markets, and U.S. economic reaction	6,594	38.4%	-0.383
Reddit	Iran–U.S. geopolitical conflict and war risk	5,694	33.2%	-0.536
Reddit	Strait of Hormuz and oil-market disruption	4,874	28.4%	-0.285
YouTube	Iran–U.S. conflict, Israel, and military escalation	10,294	39.4%	-0.416
YouTube	Global power politics and U.S.–China rivalry	8,384	32.1%	-0.290
YouTube	Oil, gas prices, and global energy supply	7,458	28.5%	-0.307

The main topic result is that Reddit distributes attention fairly evenly across market/presidential reaction, geopolitical conflict, and Hormuz/oil-market disruption. YouTube is more concentrated in Iran–U.S./Israel military-escalation framing, although global-power and energy-supply frames remain substantial. This suggests that both platforms discuss the same crisis, but they organise it through different emphases: Reddit connects the Strait of Hormuz to market reaction and policy debate, while YouTube more often frames it through military escalation and broadcast-style geopolitical commentary.

The sentiment column in Table 5 also shows that not all topics carry the same emotional tone. On Reddit, the Iran–U.S. geopolitical conflict frame is the most negative, followed by the Trump/markets frame, while the direct Strait of Hormuz and oil-market disruption frame is less negative on average. On YouTube, the military-escalation frame is the most negative, while the global-power and oil/gas supply frames are somewhat less negative. This pattern suggests that the strongest negativity comes from escalation and security risk, while market and energy discussion is somewhat more analytical or speculative.

The cross-platform topic association is statistically clear but modest. The shared-topic chi-square test is significant ($p < 0.001$), but Cramer's $V = 0.0876$. Therefore, platform and topic are associated, but not strongly enough to claim separate discourses. The better conclusion is that Reddit and YouTube share a common crisis vocabulary around Hormuz, Iran, oil, shipping, prices, and escalation, while weighting those frames differently.

Lastly, topic stability also limits over-interpretation. The mean seed Jaccard is 0.351 for Reddit and 0.438 for YouTube, so the LDA outputs should be treated as broad frames rather than exact categories. This is why topics are used to interpret platform-level patterns, not to make strong claims about the precise topic of every individual comment.

5 Network and Community Results

5.1 Network definition and shape

The author node, directed weighted reply edge network is the primary network for this project because it captures direct user to user interaction: who replies to whom, how often, and in which platform context. It is therefore a conversation graph. This definition is appropriate because the project asks how communities frame the Strait of Hormuz crisis and how influence, community structure, and diffusion differ across platforms.

Table 6 shows that Reddit and YouTube produce structurally different reply networks. Reddit has 7,938 author nodes and 10,668 collapsed directed reply edges, with 88.9% of nodes in the largest weakly connected component and reciprocity of 28.2%. This indicates a connected and interactive discussion space. YouTube has 5,789 author nodes and 5,229 edges, but only 45.0% of nodes are in the largest weakly connected component and reciprocity is almost zero. This reflects YouTube's platform structure: many comments attach to videos or top-level comments without developing sustained author-to-author conversation. Figure 2 illustrates this contrast visually: Reddit has a denser central conversation core, while YouTube breaks into smaller reply clusters around channels and videos. The PageRank curves below the network displays show that both platforms have a small high-attention head, but the structural meaning differs because Reddit's high-PageRank authors sit inside a more connected and reciprocal conversation graph.

Table 6: Network structure summary.

Platform	Nodes	Edges	Largest WCC	Reciprocity	Clustering	Modularity	Top 1% PR
Reddit	7,938	10,668	88.9%	28.2%	0.0239	0.902	17.9%
YouTube	5,789	5,229	45.0%	0.08%	0.0046	0.971	12.9%

5.2 Influence and robustness

Influence is interpreted as a structural role, not as a definitive ranking of individual users. Reddit shows the clearer hub pattern: the top 1% of authors hold 17.9% of PageRank mass, compared with 12.9% on YouTube (Table 6). This indicates that attention in Reddit reply chains is more concentrated around a small set of highly visible authors.

The robustness checks qualify this result. As shown in Figure 3, Reddit's exact PageRank ranking is only moderately stable under edge resampling, with a top-20 overlap of 45.5%. Therefore, the robust finding is not the precise identity or order of the top users, but the presence of a concentrated hub-like interaction structure. YouTube has a higher top-20 overlap of 76.3%, but this stability occurs in a much shallower and less reciprocal reply network, so it should not be interpreted as stronger conversational influence.

Community assignments are more reliable than individual influence rankings. The Louvain partitions are stable across seeds, especially on YouTube (ARI = 0.938) and reasonably stable on Reddit (ARI = 0.743), as reported in Figure 3. This supports using communities as downstream units for sentiment and topic analysis, while keeping the influence interpretation role-based: Reddit contains discussion hubs and reciprocal responders, whereas YouTube mainly contains fragmented reply participants around videos and channels.

5.3 Community interpretation

Louvain returns 355 raw communities on Reddit and 833 on YouTube, but these counts should not be read as the number of substantive publics. As suggested by the high modularity values in Table 6, both reply graphs are strongly segmented. Much of this segmentation follows platform-specific organisational units: Louvain purity is 0.832 against subreddit scope on Reddit and 0.845 against channel/video scope on YouTube. Community analysis therefore adds interaction structure, but it does not replace subreddit, channel, or video scope analysis.

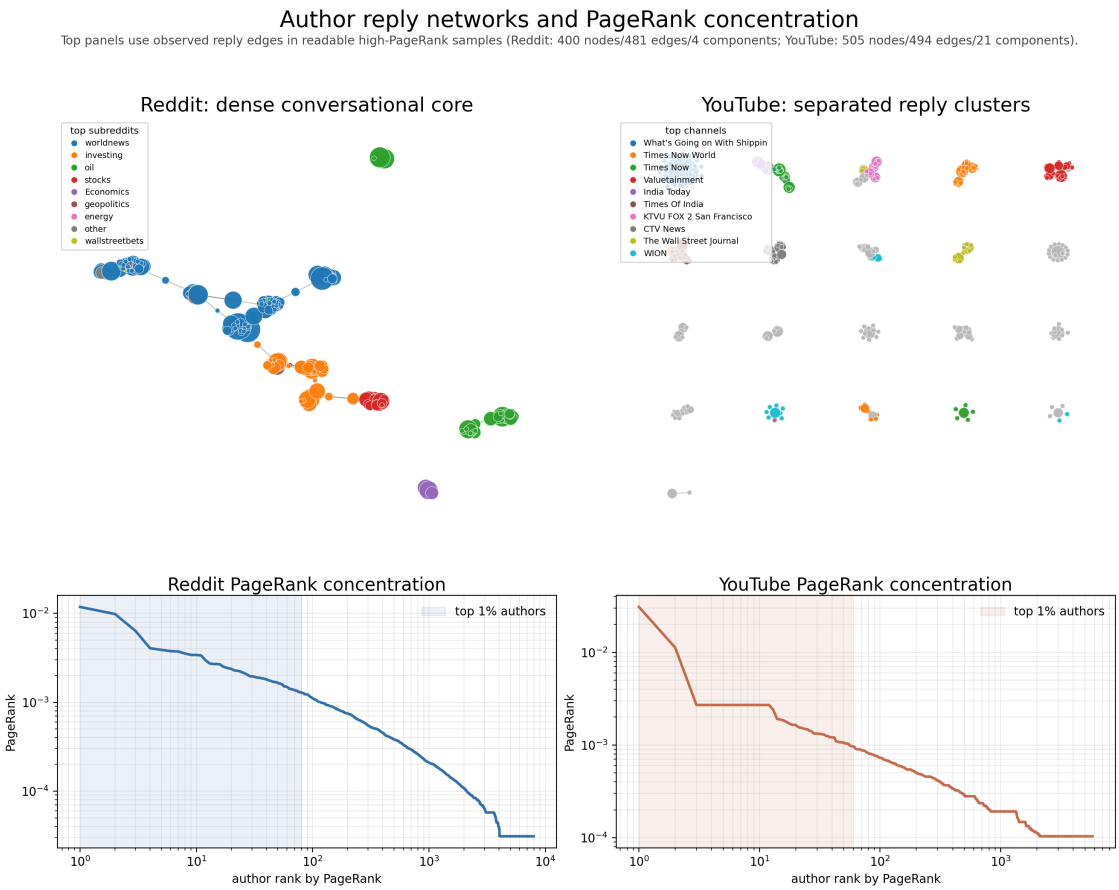

Figure 2: Reply-network display and PageRank concentration. The top panels show readable high-PageRank samples with observed reply edges: Reddit is shown as a dense conversation core, while YouTube is shown with separated reply components to preserve its fragmentation. The lower panels show how PageRank attention is distributed across authors.

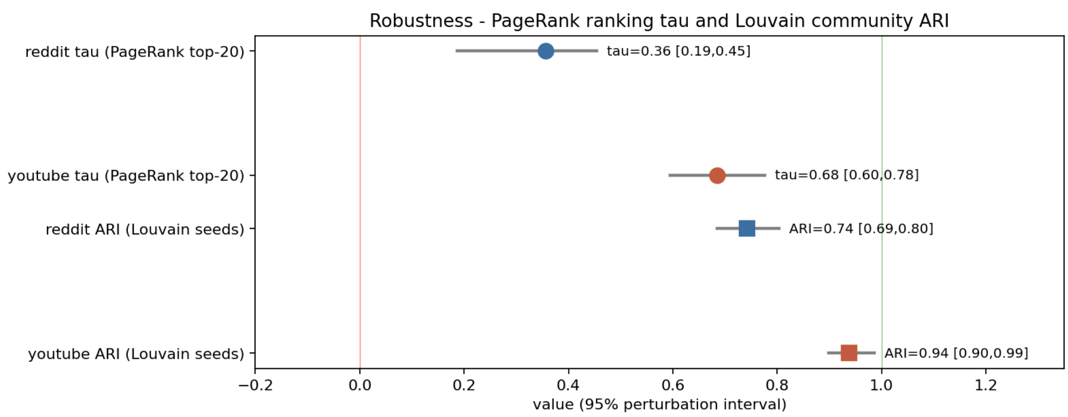

Figure 3: Robustness checks for influence and communities. Community partitions are more stable than exact Reddit PageRank rankings.

For this reason, community level NLP is interpreted only after coverage and size filtering. Figure 4 shows the main limitation: Reddit has much stronger community coverage than YouTube, with 81.1% of comments joined to author communities, compared with 41.0% on YouTube. Reddit community results can therefore be interpreted more broadly, while YouTube community results describe only the subset of users who participate in reply-network interaction.

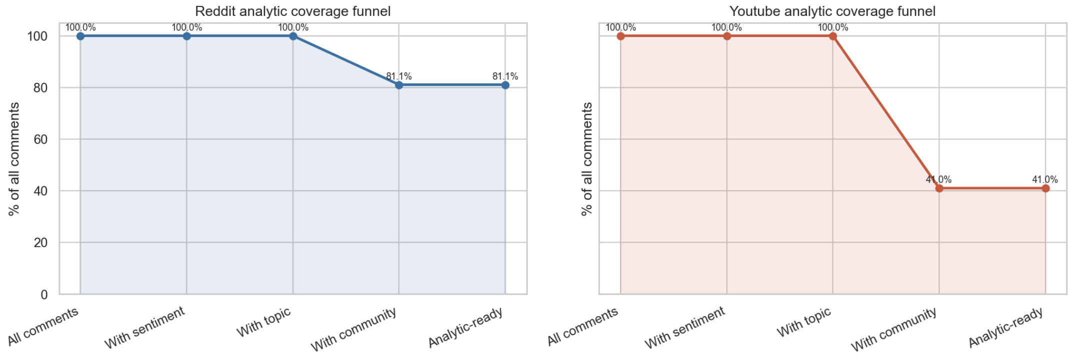

Figure 4: Coverage gates for community-level NLP. Reddit community findings are stronger; YouTube community findings are conditional on reply-network participation.

The threshold is 100 comments per community. After this gate, Reddit retains 40 communities covering 88.4% of assigned Reddit comments, while YouTube retains 28 communities covering 52.8% of assigned YouTube reply-network comments. This filtering step turns the large raw Louvain partition into a smaller and more reliable set of interpretable communities, without changing the full graph used for network metrics, PageRank, or diffusion analysis.

Figure 5 maps these communities by sentiment, topic, dominant scope, and comment volume. The main pattern is that Reddit contains several sizeable, negative communities around geopolitics, energy, world news, and investing, while YouTube has fewer reply-network communities. Table 7 gives concrete examples: the Reddit communities are mostly negative and topic-specific, whereas the largest YouTube example is a shipping/oil reply community whose interpretation remains conditional on partial coverage.

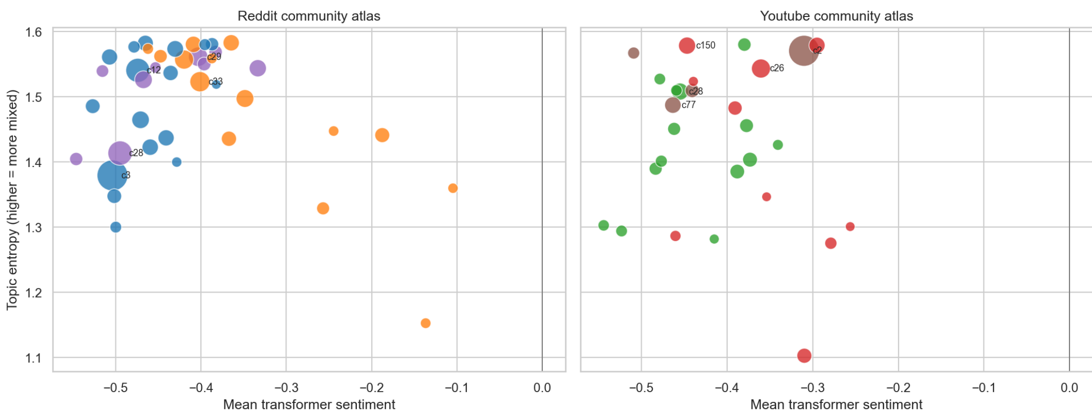

Figure 5: Threshold communities. Bubble size reflects comment volume; position and colour summarise sentiment, topic, and dominant scope. The atlas is interpreted after the 100-comment gate, not from raw Louvain counts.

Overall, the community results should be read through the subset shown in Figure 5 and Table 7, not through the raw Louvain counts. The key finding is that communities are

Table 7: Examples of interpretable report-grade communities.

Platform	Scope	Comments	Authors	Neg.	Interpretation
Reddit	geopolitics	1,232	473	66.6%	Hormuz/oil-disruption: negotiation, shipping law, escalation risk.
Reddit	worldnews	762	506	67.1%	Security debate with heavy Trump/market reaction framing.
Reddit	energy	737	318	64.2%	Energy-security community linking Hormuz disruption to war risk.
Reddit	investing	514	274	52.7%	Investor community focused on hedging, airlines, and fuel-market.
YouTube	shipping channel	803	337	46.2%	Large shipping/oil reply community.

useful for sentiment and topic summaries once filtered, but Reddit provides the stronger basis for community interpretation because its reply network coverage is much higher.

6 Diffusion Results

Diffusion is analysed at the unit supported by each platform. Reddit has nested submission trees, so submissions can be treated as cascades. YouTube mostly has replies to top-level comments, so it is interpreted as shallow engagement rather than as an equivalent cascade network. This keeps the comparison aligned with platform structure rather than forcing the two platforms into the same diffusion model.

6.1 Reddit cascade shape and timing

The Reddit data support cascade analysis. Across 55 submission cascades, the median cascade contains 305 comments, reaches maximum depth 9, and receives its first reply after 176 seconds, or about 2.9 minutes. Half of cascade comments arrive within a median of 3.2 hours, showing that Reddit discussion develops quickly but still remains threaded. Figure 6 therefore supports the interpretation of Reddit diffusion as both wide and deep.

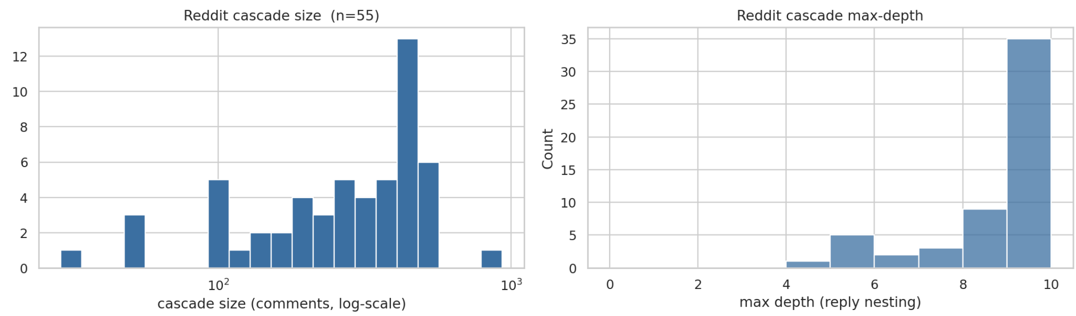

Figure 6: Reddit cascade size and depth. The Reddit data supports cascade analysis because submissions produce large, nested reply trees rather than only isolated top-level reactions.

Cascade size differs by subreddit. The largest median cascades appear in `worldnews` (489.5 comments), `oil` (477.0), and `stocks` (460.0), while smaller cascades appear in `geopolitics` (181.5) and `Economics` (146.5). This suggests that diffusion reflects subreddit scale, posting norms, and news salience, not only interest in the Strait of Hormuz crisis.

6.2 Sentiment, topic, and spread

Cascade spread is not explained by negativity alone. The most negative sentiment tercile has a smaller median size (187 comments) than the middle and least-negative terciles (362 and 390 comments). The most negative cascades are still deep, but they are not the largest. This means that outrage alone does not explain attention to the crisis.

Topic frame is more informative, as shown in Figure 7, but it should be read with the same caution as the topic model itself. Using the refreshed topic assignments, Strait of Hormuz and oil-market disruption cascades have the largest median size (376.0 comments), closely followed by Trump/markets/U.S. economic reaction cascades (351.0). Iran–U.S. geopolitical conflict cascades are smaller by median size (252.5), but they are deeper and have the highest median negative share (64.1%). The topic evidence therefore suggests that wider spread is not simply the same thing as stronger negativity.

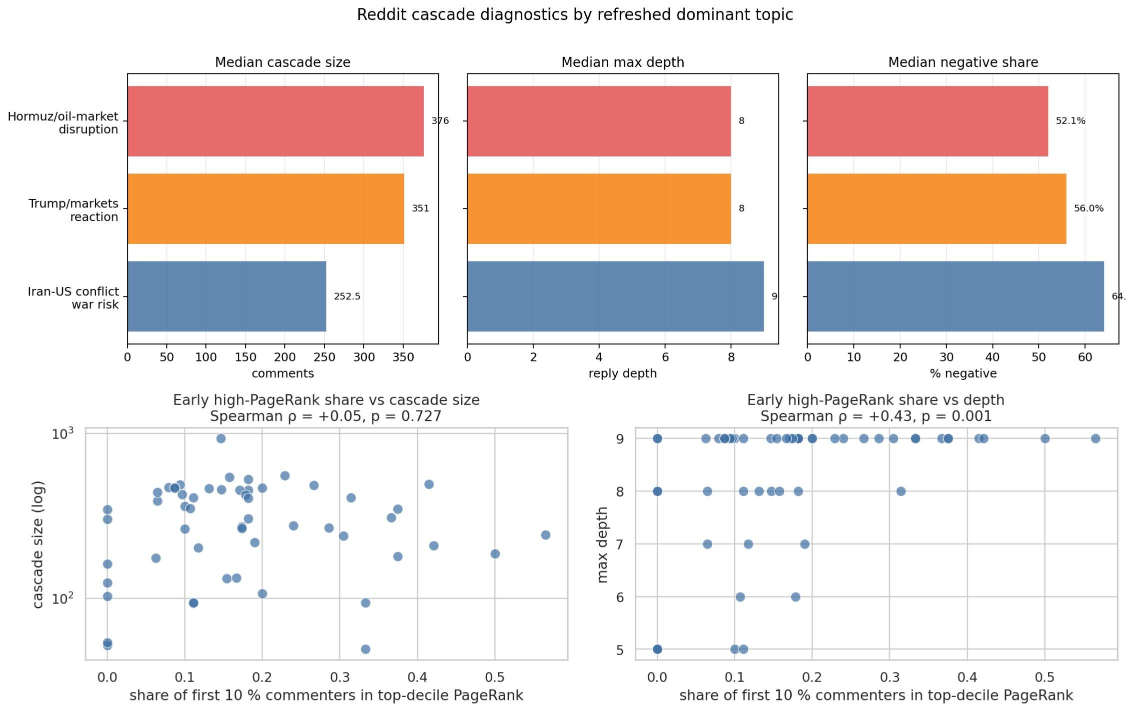

Figure 7: Additional Reddit diffusion diagnostics. Refreshed topic labels are linked to cascade size, depth, and negative share, while early high-PageRank participation is associated with depth rather than total cascade size.

6.3 Influence and independent-cascade simulation

Influence and diffusion are connected only in a limited way. Early participation by high-PageRank authors is not associated with larger cascades, but it is associated with deeper cascades. As shown in Figure 7, central users appear to sustain threaded discussion rather than simply increase total cascade size.

The Independent Cascade Model gives the same cautious message. At $k = 20$, betweenness seeds produce the highest mean expected spread (27.7 nodes), followed by PageRank seeds (23.6) and random seeds (21.4). Centrality based seeding can therefore outperform random seeding, but PageRank is not the single best diffusion strategy.

6.4 YouTube engagement

YouTube is not comparable to Reddit as a cascade network. Out of 18,012 top-level comments, only 12.36% receive any reply, and engaged comments typically receive just one reply. The median first reply arrives after about 2.1 hours, much slower than Reddit's 2.9-minute median. YouTube therefore adds breadth of public reaction, but its reply structure provides limited evidence of deep diffusion.

Overall, diffusion should be interpreted using the full Reddit cascade structure and the full

YouTube top-level engagement structure. Community reduction is useful for sentiment and topic summaries, but it is not the correct unit for measuring diffusion.

7 Platform Comparison and External Context

7.1 Cross-platform synthesis

The main cross platform finding is asymmetry: YouTube contributes more scale, while Reddit contributes deeper interaction. Figure 8 summarises this contrast. Sentiment and topic differences exist, but they are modest and the stronger difference is structural. Reddit has a more connected and reciprocal reply network, while YouTube is larger but more fragmented and more dependent on video/channel context.

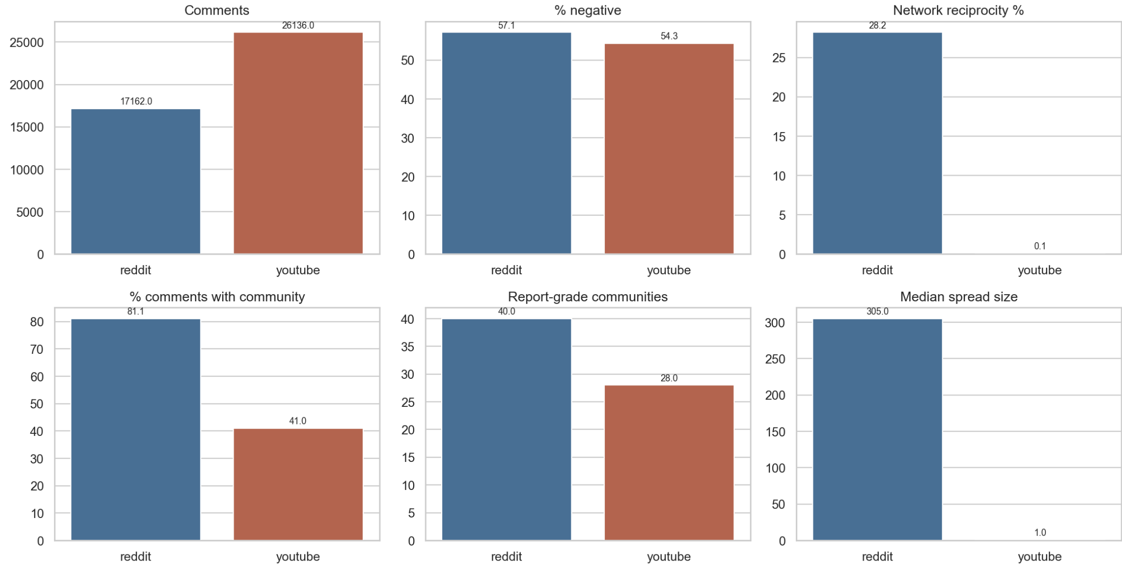

Figure 8: Compact platform comparison. The strongest contrast is structural, not simply emotional: YouTube is larger, Reddit is more conversational.

Table 8: Key cross-platform statistical results.

Comparison	Effect	95% CI	Interpretation
Mean sentiment (R–YT)	-0.061	[-0.071, -0.052]	Reddit is slightly more negative.
Rank-biserial sentiment	-0.035	[-0.046, -0.024]	Score distributions barely separate.
Sentiment labels, Cramer's V	0.078	–	Association is clear but small.
Combined topic, Cramer's V	0.088	–	Topic mix differs, but broad crisis vocabulary is shared.
Reddit PageRank, Kendall's τ	0.356	[0.187, 0.454]	Exact top-influencer ranks should be treated cautiously.
Reddit Louvain ARI	0.743	[0.685, 0.804]	Communities are stable enough for thresholded summaries.

Table 8 clarifies the size of the differences. Reddit is slightly more negative on average, but the rank-biserial effect is only -0.035, so the sentiment distributions are barely separated. Topic mix also differs only modestly ($V = 0.088$), meaning both platforms share the same broad crisis vocabulary while weighting frames differently. By contrast, the network evidence is more consequential: Reddit supports community and diffusion interpretation better, although exact PageRank rankings remain unstable and should not be over-interpreted.

Substantively, both platforms frame the Strait of Hormuz crisis as a geopolitical and energy-market risk. Reddit turns this into threaded debate about responsibility, shipping law, market pricing, airline exposure, and fuel hedging. YouTube turns the same event into broader audience reaction around videos. The answer to the research question is therefore not that one platform is simply more negative, but that the same crisis frame is organised through different interaction systems.

7.2 External market context

The external price analysis is used as temporal context, not as causal validation. Figure 9 aligns daily sentiment with FRED energy price series, showing that discussion and prices moved within the same crisis window. The clearest reportable correlations are based on differenced series, because raw correlations may reflect shared trends.

The strongest useful signals are mixed rather than causal. Reddit discussion volume is negatively associated with later jet-fuel price changes ($r = -0.410$, lag $+7$) and positively associated with Brent changes at an earlier lag ($r = 0.403$, lag -4). Negative-sentiment changes also show some association with WTI changes ($r = 0.322$, lag -2). These results suggest that social media attention and energy prices moved together during the crisis window, but the changing signs and lags prevent a causal interpretation.

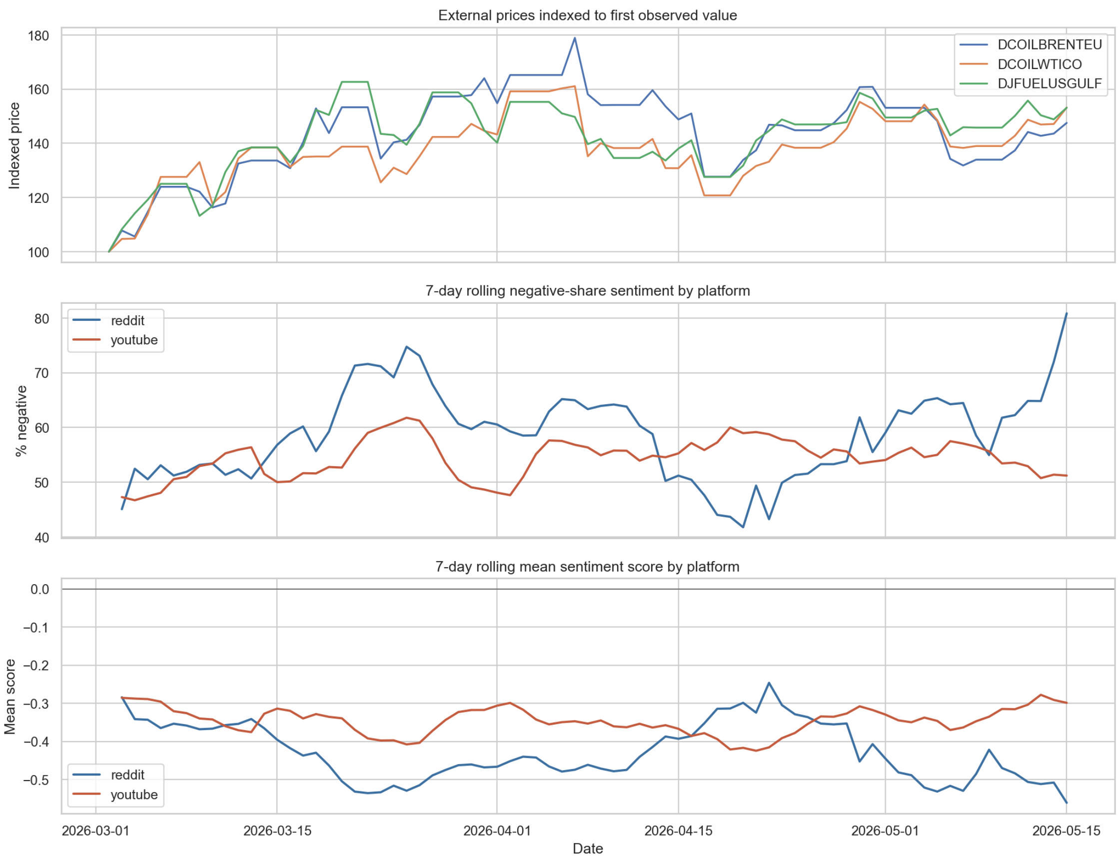

Figure 9: Daily sentiment and external price context. The figure supports temporal comparison, not causal claims.

Overall, the market context should be read cautiously. The window is short, prices are business day series while comments are daily, and multiple lags are tested. The defensible conclusion is that the Hormuz discussion occurred alongside visible energy market movement, not that social media discussion caused price changes or predicted them reliably.

8 Discussion and Conclusion

The analysis answers the research question through one main conclusion: Reddit and YouTube frame the Strait of Hormuz crisis through a shared crisis vocabulary, but organise that discussion through different interaction systems. The difference is therefore more structural than emotional.

1. The crisis is framed as a geopolitical chokepoint and energy-market risk, not as a narrow aviation fuel topic. Direct jet fuel or aviation language is rare, appearing in only 1.10% of Reddit comments and 1.70% of YouTube comments. The link to jet fuel is therefore indirect: users mainly discuss oil supply, shipping disruption, tanker movement, gas prices, Brent/WTI expectations, airline exposure, and hedging. For this reason, Hormuz is best interpreted as an upstream fuel-market risk rather than a topic discussed mainly in explicit jet fuel terms.

2. The platforms share crisis frames, but Reddit turns them into deeper interaction. Figure 8 and Table 8 show that the NLP differences are real but small: Reddit is slightly more negative, but the rank-biserial effect is only -0.035, and the topic association is modest. Both platforms discuss Iran, war risk, oil, shipping, and prices. The stronger contrast is structural: Reddit supports threaded debate about responsibility, markets, fuel exposure, and investment risk, while YouTube mainly supports broader audience reaction around geopolitical and military videos.

3. The author-reply network is valid, but interpretation must use filtered communities. The network definition is appropriate because the project studies interaction, influence, and community structure. However, the raw Louvain outputs (355 Reddit communities and 833 YouTube communities) are too fine-grained to treat as substantive publics. As shown in Table 6 and Figure 4, the defensible approach is to use the full partition for joining authors to comments, then interpret only report-grade communities. The 100-comment gate retains 40 Reddit and 28 YouTube communities, giving a smaller and more reliable basis for sentiment and topic summaries.

4. Diffusion is meaningful only at the correct platform unit. Reddit supports cascade analysis because its 55 submission trees are large and nested, with a median cascade size of 305 comments and depth 9. Figure 7 shows that spread is not driven by negativity alone: Trump/markets cascades are the widest by median size, while direct Hormuz/oil-disruption cascades are smaller but highly negative and still deep. YouTube should instead be read as shallow engagement evidence: only 12.36% of top-level comments receive any reply, so it is not equivalent to Reddit as a cascade network.

Overall, Reddit provides the stronger basis for network, community, and diffusion analysis. It has 7,938 author nodes, 10,668 reply edges, an 88.9% largest weakly connected component, and 28.2% reciprocity (Table 6). YouTube adds scale, with 26,136 comments and 19,676 authors, but its reply structure is fragmented and shallow. The external price context in Figure 9 supports only a temporal comparison with energy markets, not causal claims. The final answer is therefore that Reddit turns the Hormuz crisis into threaded debate about market risk, U.S./Iran responsibility, oil supply, shipping, and investment exposure, while YouTube turns the same crisis into broader but shallower public reaction around videos.

9 Limitations

The study has five important limitations.

Sampling. Reddit is a curated set of high-relevance threads from eight subreddits, and YouTube is an API-search sample from six topical queries. The executed analysis corpus contains 17,162 Reddit comments and 26,136 YouTube comments, but the findings describe only the collected corpus, not all public discussion of the crisis.

Platform comparability. Reddit and YouTube do not expose equivalent conversation structures. Reddit supports deep threaded replies; YouTube mainly supports top-level commentary and shallow replies. Network metrics are therefore compared as platform structures, not as direct measures of which public is more engaged.

Model uncertainty. Transformer sentiment, VADER, lexicon counts, and topic models are automated approximations. The agreement between VADER and the transformer is moderate (56.16%, Cohen's kappa 0.327), and LDA seed stability is limited, especially for Reddit.

Community scope and coverage. Louvain communities are partly shaped by subreddit, channel, and video boundaries. They are useful for joining network structure to text, but raw community counts are not substantive group counts. Community-level YouTube findings are especially conditional because the reply-network join covers 41.0% of YouTube comments, compared with 81.1% of Reddit comments.

External validation. Price/discussion correlations are exploratory. The market window is short, many lags are tested, and correlation is not causation. The external series help contextualise timing, but they do not validate the social-media models in a predictive or causal sense. The FRED jet-fuel series is also a U.S. Gulf Coast spot proxy rather than a direct measure of every global aviation fuel market.

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A Reproducibility Appendix

A.1 Repository workflow

The repository is organised as a small collector layer plus eight self-contained notebooks, executed in numeric order: `01_youtube_preprocessing.ipynb`, `02_reddit_preprocessing.ipynb`, `03_basic_nlp.ipynb`, `04_network_analysis.ipynb`, `05_topics_sentiment.ipynb`, `06_diffusion_analysis.ipynb`, `07_community_sentiment_topics.ipynb`, and `08_platform_comparison.ipynb`. Outputs are written under `data/processed/` and `plots/`.

A.2 Representative sample

The submitted data sample is under `data/sample/`. It contains anonymised comment rows with NLP/community fields and sampled Reddit/YouTube reply edges. This sample is sufficient to show the structure of the network and text data while remaining below the 10 MB assignment limit.

A.3 Dependencies

The main dependencies are Python, pandas, NumPy, scikit-learn, NetworkX, python-louvain, VADER, matplotlib/seaborn, langdetect, optional transformers/PyTorch, `requests`, `python-dotenv`, and the YouTube API client. Exact instructions are in the repository `README.md`.